STRULE AUTOMATION
Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Why Is the Crusher Tripping?

Who this is for and what it does: a bench reference for the maintenance tech, plant electrician, or operator working a crushing and screening circuit that keeps dropping out. It helps you separate a real protective trip from a nuisance one, trace which upstream or downstream condition actually pulled the circuit offline, and decide what to check before you call anyone.

How to use this

Keep it open at the panel or on your phone. When the circuit trips, resist the urge to just hit reset and walk away. Work the symptom you actually have, write down what the human-machine interface (HMI) or programmable logic controller (PLC) showed at the moment of trip, and follow the checks. The goal is a cause, not just a machine that runs until it trips again in twenty minutes.

A trip is information. The protection did its job and told you something. A "nuisance" trip is not a lie. It usually means the threshold is set tighter than the real process needs, or a sensor is drifting, or a genuine fault is small enough that it looks like noise. Your job is to find out which.

First move: read the first-out, not the last alarm

Most modern crusher control systems latch a first-out or first-fault record: the condition that fired first, before the cascade. This is the single most useful piece of data you have. When a crusher stops, everything downstream loses feed and everything upstream backs up, so within a second or two you can have ten alarms standing. Nine of them are consequences. One is the cause.

- Find the first-out timestamp on the HMI alarm history. Sort by time, oldest first in the trip window.
- If there is no first-out logic, look at the alarm with the earliest millisecond timestamp and treat downstream conveyor and feeder stops as secondary until proven otherwise.
- Log the exact trip value if the system captures it (motor amps, lube pressure in pounds per square inch, cavity level percent). "High amps" is a story; "412 A on a 380 A rated motor for 3 seconds" is evidence.

If you take nothing else from this guide: chase the first event, not the loudest one.

Symptom → likely causes → what to check / what to log

Crusher drive trips under load, motor amps spike then drop

Likely cause	What to check	What to log
Genuine mechanical overload (packed chamber, wet sticky feed, tramp iron)	Chamber for bridged or packed material; feed moisture; last few feed samples	Peak amps, ramp shape, feed rate at trip, material condition
Overload relay or drive current limit set too conservative for actual duty	Rated full-load amps (FLA) on nameplate vs. trip setpoint and class; drive current-limit parameter	Setpoint, trip class/time, nameplate FLA, typical running amps
Drive fault (overcurrent, DC-bus, or insulated-gate bipolar transistor / IGBT) rather than process overload	Drive fault code, not just the PLC's generic "crusher fault"	Exact drive fault code and timestamp
Worn liners changing the crush and drawing more power	Liner wear measurement / hours since change	Liner hours, closed-side setting

A true overload builds. Amps climb as material packs. A drive electrical fault often trips clean with no matching process reason. That difference tells you whether to look at the chamber or the drive.

Tramp metal detector or metal-in-feed trip

Likely cause	What to check	What to log
Real tramp iron (bucket tooth, ground-engaging tool, bar)	Feed belt at the detector; magnet condition upstream	What was found, where, size
Detector sensitivity set too high, catching wet, highly conductive or mineralized fines (magnetite/ironstone), i.e. product effect	Sensitivity setting vs. commissioning baseline; recent changes	Setting, feed moisture, false-trip frequency
Detector head or coil fault, marginal wiring, moisture in a junction box	Detector self-test/diagnostics; cable and connector condition	Fault flags, resistance/insulation readings if taken

Treat this one with respect. A tramp metal trip protects the crushing chamber and the liners from a strike that can cost you far more than the downtime. Do not desensitize a metal detector to stop nuisance trips until you have physically confirmed there is no metal and no wiring fault. Chase the annoyance instead of the cause and you can let a strike straight into the mantle.

One more thing on cone crushers: a tramp-metal detector that stops the feed is an electrical trip, but an uncrushable that reaches the chamber is usually handled mechanically by the hydraulic tramp-relief system. The accumulators dump, the ring lifts to pass the iron, then resets. That can show up as ring bounce or repeated relief events rather than an electrical trip in the alarm log. If the "trips" you are chasing are really relief events, you are looking in the wrong system: check the tramp-relief accumulator pre-charge and the relief and reset pressures, not the detector.

Lube or hydraulic pressure trip (crusher won't stay running / won't start)

Likely cause	What to check	What to log
Genuine low lube oil pressure or flow	Pump running? Pressure at gauge vs. transmitter reading; oil level; filter differential	Pressure trace, oil temp, filter differential pressure (dP)
Cold oil, high viscosity at startup in cold weather	Oil temperature vs. minimum start-permit temp; heater working	Oil temp at trip, ambient
Pressure/temperature transmitter drift or a marginal switch	Compare transmitter to a known-good gauge at the same port	Transmitter reading vs. gauge, deviation
Return-line restriction or flow-switch sticking	Return flow, flow-switch actuation	Flow reading, switch state

Lube protection guards the bearings and eccentric, the expensive heart of the machine. If the pressure is genuinely low, the trip is correct and the machine should stay down. If the transmitter is lying, fix the transmitter; do not raise the trip window to ride through a reading you don't trust.

High bearing or oil temperature trip

Likely cause	What to check	What to log
Genuine overheating or loss of cooling	Oil cooler and cooling fan running; coolant or air flow; oil level and circulation	Temperature trace, rate of rise, ambient, cooler/fan status
Degraded or wrong-grade oil losing its film strength	Oil condition (color, smell, last sample/change); grade against spec	Oil hours since change, last analysis, grade in use
Resistance temperature detector (RTD) or thermocouple drift, or an open sensor reading high	Compare the RTD or thermocouple to a contact thermometer at the housing; check for an open circuit	Sensor reading vs. contact reading, deviation

Likely cause	What to check	What to log
Cooler or fan fault, blocked or fouled cooler	Cooler cleanliness, fan operation, thermostatic valve position	Cooler status, fan state, valve position

This one behaves differently from a pressure trip. Low lube pressure drops the machine out in an instant; a bearing or oil temperature trip climbs slowly, over minutes, as heat builds. Use that slow ramp: trend the rise and you can usually tell a real thermal problem (a steady climb toward the limit, often with a matching loss of cooling) from a sensor going bad (a step change, or a reading that disagrees with a contact thermometer on the same housing). A countershaft bushing RTD reading high while the main or eccentric bearing thermocouple sits normal points at one bushing or one sensor, not the whole oil system. If the temperature is genuinely high, the trip is doing its job and the machine should stay down until you find the heat source.

Cavity / chamber level trip (high or low)

Likely cause	What to check	What to log
Real over-feed choking the chamber (high level)	Feeder speed control, feed-rate loop tuning, level sensor sightline	Level trace, feeder speed, feed control mode
Real starvation / loss of feed (low level), often an upstream stop	Upstream feeder/conveyor status at trip time	Upstream device states, sequence
Ultrasonic or radar level sensor seeing dust, buildup, or splash	Sensor face condition, dust conditions, echo quality	Sensor diagnostics, signal quality
Level trip band set tighter than the process naturally swings	Trip band vs. normal operating variation	Setpoint band, observed swing

High-level trips are frequently a feed-control tuning problem rather than a crusher problem. If the feeder loop is hunting, the level overshoots and clips the trip on the peaks.

Downstream conveyor or feeder dropped and pulled the crusher offline

Likely cause	What to check	What to log
Belt-drift / tracking switch tripped on a wandering belt	Belt tracking at head, tail, and load point; drift-switch actuation and mounting position	Which switch, belt loading at trip, tracking condition
Belt slip (drive slip, plugged belt, or overloaded conveyor)	Zero-speed / belt-speed switch versus drive running; material buildup, pulley lagging, takeup tension	Belt speed vs. drive state, slip timestamp

Likely cause	What to check	What to log
Pull-cord pulled or out of tension along the run	Pull-cord stations and tension switches on the conveyor or feeder	Which station/switch, latched state
Conveyor or feeder motor overload / drive fault	Overload relay or drive fault code on that unit; running amps vs. nameplate FLA	Fault code, trip amps, nameplate FLA

When the crusher shows a high-cavity or over-feed trip, the true first-out is often on the machine that stopped taking its product. A discharge conveyor or feeder that drops on its own drift, slip, pull-cord, or overload backs the chamber up in seconds, and the crusher trip is only the consequence. Read that unit's own first-out, not just the crusher's alarm. One caveat: a pull-cord is a safety function, not a nuisance-trip candidate. Check and repair it per the E-stop and safety chain note below; never defeat it to keep running.

E-stop or safety chain trip

Likely cause	What to check	What to log
Someone hit an e-stop, or a pull-cord on a conveyor was pulled	Every e-stop and pull-cord station in the chain; latched indicators	Which station, timestamp
Pull-cord tension out of range (too slack or too tight) after weather or vibration	Cord tension switches along the run	Which switch, tension state
Safety relay / guard interlock fault or wiring open	Safety relay diagnostics, guard switch continuity	Relay fault, circuit node

This is the one place where the rule is absolute. The safety chain is not a nuisance-trip candidate to be tuned out. A loose pull-cord switch gets adjusted to spec or replaced; it never gets jumpered, bridged, or defeated to keep the plant running. If a guard or e-stop circuit is dropping the crusher, find the open and restore it properly.

Tracing which condition actually pulled the circuit offline

Crushing and screening runs as an interlocked sequence: you don't want to keep feeding a stopped crusher, or run a conveyor into a stopped screen. That same interlocking means a single fault propagates fast, so the machine that shows "tripped" is often not where the problem started. Work it like this:

1. **Start at the first-out.** Establish the earliest genuine fault, as above.
2. **Ask: upstream or downstream of that device?** If the crusher tripped on high cavity level, the cause is usually downstream: the discharge conveyor or screen stopped, product stopped moving,

the chamber backed up. If it tripped on low level or loss of feed, look upstream, where the feeder or surge pile feed stopped first.

3. **Follow the interlock logic, not your assumptions.** Pull up the PLC's permissive or interlock rung for the tripped device. It will list, in plain terms, every condition that must be true for it to run. One of those went false first. That is your thread to pull.
4. **Separate "commanded stop" from "fault trip."** A downstream conveyor that stopped on its own fault (belt slip, pull-cord, motor overload) will have its own first-out. A conveyor that stopped because the sequence told it to is just following orders. The HMI event log usually distinguishes these; learn which is which on your system.
5. **Confirm at the field device.** Once the logic points somewhere, go verify it in the real world: the gauge, the sensor face, the material, the belt. Control logic tells you where to look; it does not replace looking.

Retuning a conservative trip — properly

Sometimes the trip really is set tighter than the process needs, and the honest fix is to open the window a little. That is legitimate engineering when you do it deliberately and with a record. It is negligence when you do it to silence protection you haven't understood.

The line between the two:

- Know the equipment limit you are protecting. Bearing pressure minimum, bearing temperature limit, motor thermal rating, chamber design: the trip should sit inside the equipment's real limit with margin, never beyond it.
- Change one setpoint or one time delay at a time, and write down the old value, the new value, the reason, and the date. If it doesn't help, you can put it back.
- A short trip delay (letting a transient settle for a defined second or two) is often the right tool for a genuine nuisance trip, a brief amps spike on a slug of feed, for instance. Extending a delay so far that real damage can occur is not.
- Never move a lube-pressure, bearing-pressure, bearing-temperature, tramp-metal, or safety-chain trip to ride through a condition you suspect is real. If a sensor is drifting, replace or recalibrate the sensor. Do not compensate for a bad reading by widening the trip: you lose the protection and keep the bad sensor.

Retuning a conservative trip is adjusting where the alarm sits. Jumpering out protection is removing the alarm. The first can be good practice. The second, on a lube, bearing, or safety function, is how bearings and people get hurt. Keep those two things clearly separate.

10-minute triage (run this before you call anyone)

- ☐ Read the **first-out / earliest-timestamp** fault. Write it down with the time.

- ☐ Capture the **trip value and trend** if the system logs it (amps, pressure, temperature, level). Screenshot or note the numbers.
- ☐ Walk the chain: is the tripped device the **cause or a consequence**? Check upstream and downstream device states at the trip time.
- ☐ Check the obvious physical: **chamber packed?** feed wet/sticky? tramp metal on the belt? oil level and temperature?
- ☐ Compare a suspect **sensor reading to a known-good gauge or contact thermometer** at the same point.
- ☐ Confirm no **e-stop or pull-cord** is latched anywhere in the run.
- ☐ Note whether this is a **first occurrence or a repeat**, and what changed recently (feed source, liners, weather, a parameter someone touched).
- ☐ Only after the above: reset and observe. Watch whether it trips again at the same value.

If you have those eight things written down, whoever you call next, including me, can be useful in the first five minutes instead of the first hour.

Honest limits

This guide narrows the search; it does not replace eyes on the specific machine. Every plant's control logic, sensor set, and feed differ, and the same symptom can have a cause I haven't listed. Intermittent faults, the ones that clear before you reach the panel, are genuinely hard and sometimes need data logging over days to catch. Nothing here substitutes for the original equipment manufacturer's limits, your site's safety procedures, or lock-out/tag-out before anyone puts a hand near the machine. If a check here conflicts with your site rules or the equipment manual, the site rules and the manual win.

If the fault keeps getting handed around

The frustrating trips are usually the ones that fall between vendors: the drive supplier says it's the process, the crusher supplier says it's the controls, the controls person says it's a sensor, and the plant loses the hours. Because I don't carry a product line, I can look at the whole interlock chain (PLC, drive, HMI, sensors, whatever brand is installed) and trace the trip end to end without needing it to be someone's fault.

If you've worked the triage above and the crusher is still dropping out, or the cause keeps bouncing between suppliers, get in touch and we can talk it through: struleautomation.com or info@struleautomation.com

— Jonathan Gilmour, Strule Automation